Discrete Mathematics – Final Exam

2021 – Moed B

Instructions for students:

1. In the exam there are **10** short questions and **3** open questions. You must answer **all** of them (No choice).
2. Write your solutions in the attached papers only.
3. When asked to write explicitly a "set/function/relation", you must specify all the elements of the set/function/relation.
4. Use of any external aiding material is **forbidden**. For your convenience, a formula sheet is attached.
5. The duration of the exam is **three hours.**

Good luck!

Part A – Short Questions

- Write your answers clearly in the marked boxes.
- No need to explain how you got to the final answer, **unless it is explicitly asked**.

In counting questions, you can use any of the convenient notations which we used in class, **except** for the symbol $\sum$ and three dots, '…'.

Question 1 (6 points) – Simplify the definition.

Let $A = \{\emptyset, \{1\}\}$, provide an explicit representation of the following set :

$(P(A) \cap A) \times P(\{\emptyset\})$.

$\{(\emptyset, \emptyset), (\emptyset, \{\emptyset\})\}$

Question 2 (6 Points)

Given are the following three propositions :

$\alpha = p \to \neg q$, $\beta = p \leftrightarrow q$, $\gamma = \neg p \vee q$.

Provide a fourth proposition δ, such that $\frac{\alpha, \beta, \gamma}{\delta}$ forms a **sound** inference rule.

$\delta = p \to q$

Question 3 (6 points).

Let $f : X \to Y$ be a bijection.

We define $F: P(X) \to P(Y)$ as follows: $F(A) = \{f(a) \mid a \in A\}$.

Determine (no explanation required) if the function is injective, and if it is surjective.

- If it's only injective, present two <u>different</u> left inverse functions (no proof).
- If it's only surjective, present two <u>different</u> right inverse functions (no proof).
- If it's both, present its unique inverse function (no proof).

This function is a bijection. We specify its' inverse:

$F^{-1}: P(Y) \to P(X)$

$F^{-1}(B) = \{a \in X | f(a) \in B\} = \{f^{-1}(b) \mid b \in B\}$.

Question 4 (6 points).

Present the following expression as a short numeral expression (no Sigma, no '…', though your answer can include n):

$$\sum_{k=1}^{n} \binom{2n}{k} \binom{3n}{n-k} k \cdot (n-k)$$

$\binom{5n-2}{n-2} \cdot (2n) \cdot (3n)$.

The expression answers the following question : "Given a class of 5n students – 3n of which are boys, 2n of which are girls – How many ways are there to pick a committee of n children, from which we also choose one boy and one girl to be the committee directors?

Question 5 (6 points)

A pack of cards has 52 cards of 4 suits. Each suit has 13 cards, numbered 1-13.

A hand of 5 cards is a flush when all 5 cards are of the same suit, and the 5 numbers on the cards are not all consecutive (eg 1,2,3,4,6 is ok, but 1,2,3,4,5 is not).

How many possible flushes are there?

$4 \cdot (\binom{13}{5} - 9)$

We use R.O.P: Pick a suit and then pick 5 non-consecutive numbers.

The set of non-consecutive numbers that can appear on the cards is of size $\binom{13}{5} - 9$, the number of suits is 4.

Question 6 (6 points)

The IDC computer lab room has 6 computers with 6 different programming languages (each computer supports a single different programming language) : Haskell, Java,

We have 6 students listing the programming languages they are familiar with:

- Jill : C++, Java, C#.
- Gabbie : Haskell, C++, Java, Python.
- Danny : Java, C#.
- Haylee : Haskell, C++, Java, C#, Lisp.
- Sarah : C++, C#.
- Ziggy : C++, Java.

Can we seat each student at a different computer such that each student has a computer that supports a language they are familiar with? Explain shortly.

Note that for S = {Ziggy, Sarah, Danny, Jill}, N(S) = {C++, C#, Java}.

This means |S| > |N(S)| - violating Hall's marriage theorem rule which means there does not exist a solution to this problem.

Question 7 (6 points)

We have 4 letters: A,B,C,D. For each letter we have 4 copies (Overall we have A,A,A,A,B,B,B,B,C,C,C,C,D,D,D,D). How many ways are there to create 10-letter words from these letters, such that every letter appears at least twice?

$$\binom{4}{1}\binom{10}{4,2,2,2}+\binom{4}{2}\binom{10}{3,3,2,2}$$

We use R.O.S and divide into cases : First case is one letter appears 4 times and the others appear twice. Second case is two letters appear 3 times and the others appear twice.

Question 8 (6 Points)

Provide a relation R over ℕ that is transitive, anti-reflexive, and not anti-symmetric, or explain why such a relation does not exist.

Such a relation does not exist – Assume R is transitive and not anti-symmetric. Because it is not anti-symmetric, there exist $x \neq y$ such that $xRy \wedge yRx$, Because of transitivity we can conclude xRx which means R is not anti-reflexive.

Question 9 – 6 Points.

Let $A = \{1,2,3,4,5\}$.

Count the number of equivalence relations R over A such that :

$\{(1,1),(2,1),(3,3),(3,5)\} \subseteq R$ and $|R \cap \{(1,3),(4,2)\}| \leq 1$.

4.

We divide into cases on the intersection:

If the intersection is empty : 3,5 share an equivalence class, 1,2 share a **different** equivalence class which 4 is not a member of : 2 options in this case.

If the intersection has a single element : That element could be (1,3) (and not (4,2)) in which case 1,2,3,5 all share the same equivalence class and 4 is on its' own **separate** equivalence class - 1 option.

That element could also be (4,2) in which 3,5 share an equivalence class and 1,2,4 share a **different equivalence class** – 1 option.

Question 10 (6 points)

How many ways are there to divide 80 identical balls into 5 different cells (numbered 1-5) such that no cell has more than 24 balls?

We use Inclusion/Exclusion :

$$\binom{84}{4} - \binom{5}{1} \cdot \binom{59}{4} + \binom{5}{2} \cdot \binom{34}{4} - \binom{5}{3} \cdot \binom{9}{4}$$

Part B – Open Questions

- Write your answers clearly in the marked boxes. You must prove your answers.
- **Write clear and readable answers! Your grade will also depend on the clarity of your proof and not only the correctness of your answer!**

Question 11 (10 points)

For a function $f : A \to A$, we denote:

$f^0 = i_A \quad , \; f^1 = f \quad , \quad f^n = \underbrace{f \circ f \circ f \circ \ldots \circ f}_{n\ times}$

a. Prove : If there exists a natural number k > 0 such that $f^k = i_A$, f is injective and surjective.
b. Disprove : If $f: A \to A$ is injective and surjective, then there exists a natural number $k > 0$ such that $f^k = i_A$.

a. Assume there exists a k > 0 such that $f^k = i_A$.
We prove : f is injective and surjective.
Note that $f^{k-1} \circ f^1 = f^1 \circ f^{k-1} = i_A$ (Because k > 0, we know f^{k-1} exists).
This means f is left invertible, which means it is injective. It is also right invertible which means it is surjective.
b. Define $f: \mathbb{R} \to \mathbb{R}$, by $f(x) = x + 1$.
It is plain to see f is surjective and injective.
However, for every k > 0, $f^k(x) = x + k > x$.
Therefore for every k > 0, $f \neq f^k$.

Question 12 (15 points)

We Define the sequence P(n) by :

$\forall n \in N, \quad P(n) = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{n(n+1)}$.

Prove : $\forall n \in N, \ P(n) = \frac{n}{n+1}$.

Base : For n=1- $\frac{1}{1\cdot 2} = \frac{1}{1+1}$ holds true.

Inductive assumption : Assume P(n) holds true, meaning $P(n) = \frac{1}{1*2} + \frac{1}{2*3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$.

Inductive step:

$$P(n+1) = \frac{1}{1*2} + \frac{1}{2*3} + \cdots + \frac{1}{n(n+1)} + \frac{1}{(n+1)(n+2)}$$

$$= \mathrm{P(n)} + \frac{1}{(n+1)(n+2)} =_{Assumption} \frac{n}{(n+1)} + \frac{1}{(n+1)(n+2)}$$

$$= \frac{n(n+2)+1}{(n+1)(n+2)} = \frac{(n+1)^2}{(n+1)(n+2)} = \frac{n+1}{n+2}$$

Question 13 (15 points)

Let $R \subseteq A \times A$, we define relation S on A as follows:

$xSy \Leftrightarrow (xRy) \vee (x = y)$.

Prove: If R is a strict partial order on A, then S is transitive.

You can use the following logical identity in your answer (double distributivity):

$(p \vee q) \wedge (r \vee s) \equiv (p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)$

Assume R is a strict partial order on A and let $x, y, z \in A$ s.t $xSy \wedge ySz$.

Therefore by the definition of S :

$\big(xRy \vee (x = y)\big) \wedge \big(yRz \vee (y = z)\big)$.

Use double distributivity :

$(xRy \wedge yRz) \vee \big(xRy \wedge (y = z)\big) \vee \big((x = y) \wedge yRz\big) \vee \big((x = y) \wedge (y = z)\big)$.

Therefore by the transitivity of R and by the equalities above :

$xRz \vee xRz \vee xRz \vee (x = z)$.

Therefore by the definition of S : xSz.

Discrete Mathematics – Formula sheet for the exam

Set Theory:

$$A\Delta B = (A\backslash B) \cup (B\backslash A)$$
$$A \times B = \{(a,b)|\ a \in A\ and\ b \in B\}$$
$$P(A) = \{S|S \subseteq A\}$$

$$|A \times B| = |A| \cdot |B|$$
$$|P(A)| = 2^{|A|}$$

De-Morgan:

$$\overline{B \cup C} = \bar{B} \cap \bar{C} \qquad A\backslash(B \cup C) = (A\backslash B) \cap (A\backslash C)$$
$$\overline{B \cap C} = \bar{B} \cup \bar{C} \qquad A\backslash(B \cap C) = (A\backslash B) \cup (A\backslash C)$$

Distributive:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Functions/Relations:

$$i_A = \{(a,a)|a \in A\}$$

Logic:

For all formulas α, β, γ :

De-Morgan:

$$\neg(\alpha \wedge \beta) \equiv \neg\alpha \vee \neg\beta$$
$$\neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$$

Distributive:

$$\alpha \wedge (\beta \vee \gamma) \equiv (\alpha \wedge \beta) \vee (\alpha \wedge \gamma)$$
$$\alpha \vee (\beta \wedge \gamma) \equiv (\alpha \vee \beta) \wedge (\alpha \vee \gamma)$$

Negation rules:

$$\neg\forall x(\alpha) \equiv \exists x(\neg\alpha)$$
$$\neg\exists x(\alpha) \equiv \forall x(\neg\alpha)$$

Combinatorics:

Choosing k items out of n different items:

	With repetition	Without repetition
With order	n^k	$\frac{n!}{(n-k)!}$
Without order	$\binom{k+n-1}{n-1}$	$\binom{n}{k} = \frac{n!}{(n-k)! \cdot k!}$

Binomial Formula

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} \cdot x^k y^{n-k}$$

Multinomial Coefficient:

$$\binom{n}{i_1, i_2, i_3, \dots, i_k} = \frac{n!}{i_1! \cdot i_2! \cdot i_3! \cdot \dots \cdot i_k!} \quad if \;\; i_1 + i_2 + i_3 + \cdots + i_k = n$$

Multinomial Formula:

$$(x_1 + x_2 + x_3 + \cdots + x_k)^n = \sum_{i_1+i_2+i_3+\cdots+i_k=n} \binom{n}{i_1, i_2, i_3, \dots, i_k} \cdot x_1^{i_1} \cdot x_2^{i_2} \cdot x_3^{i_3} \cdot \dots \cdot x_k^{i_k}$$

Pascal's Triangle:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

$$\binom{n}{k} = \binom{n}{n-k}$$

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$$

Inclusion-exclusion principle:

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{I \subseteq \{1,2,\dots,n\},\;\; I \neq \emptyset} (-1)^{|I|-1} \cdot \left| \bigcap_{\mathrm{i} \in \mathrm{I}} \mathrm{A_i} \right|$$

Generalized Pigeonhole Principle:

If we put m pigeons in n pigeonholes, then there is at least one pigeonhole in which there will be at least $\left\lceil \frac{m}{n} \right\rceil$ pigeons.